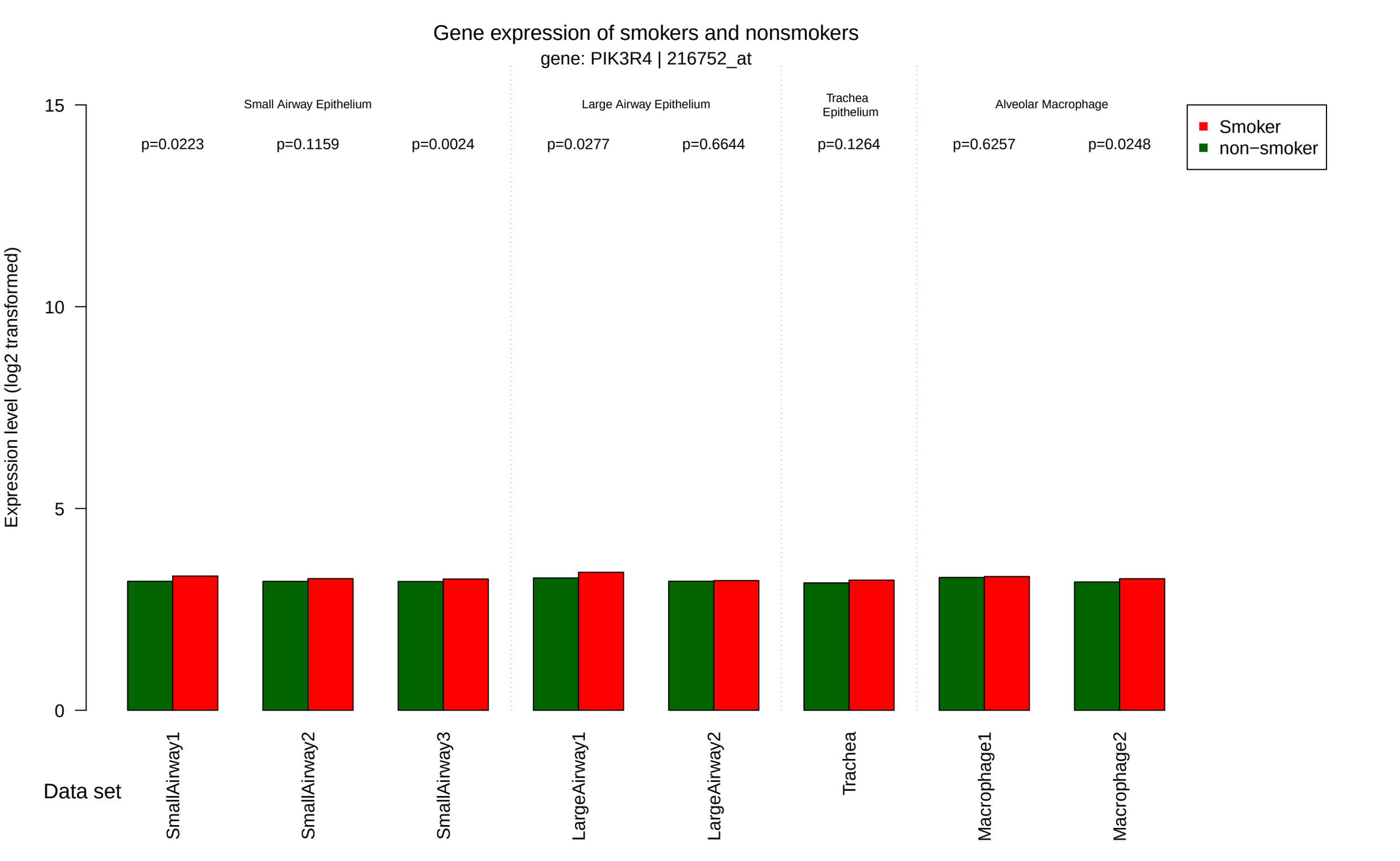
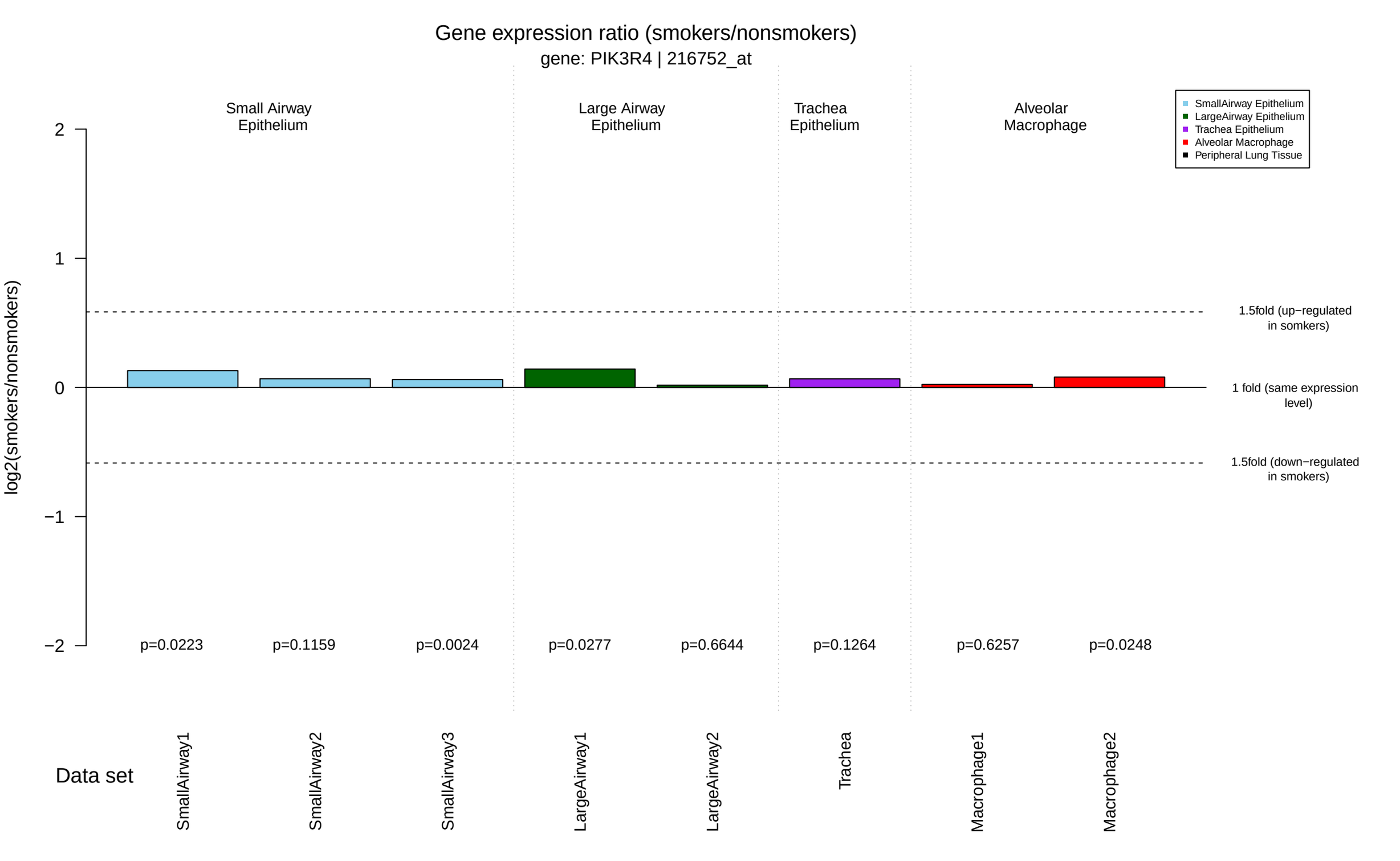
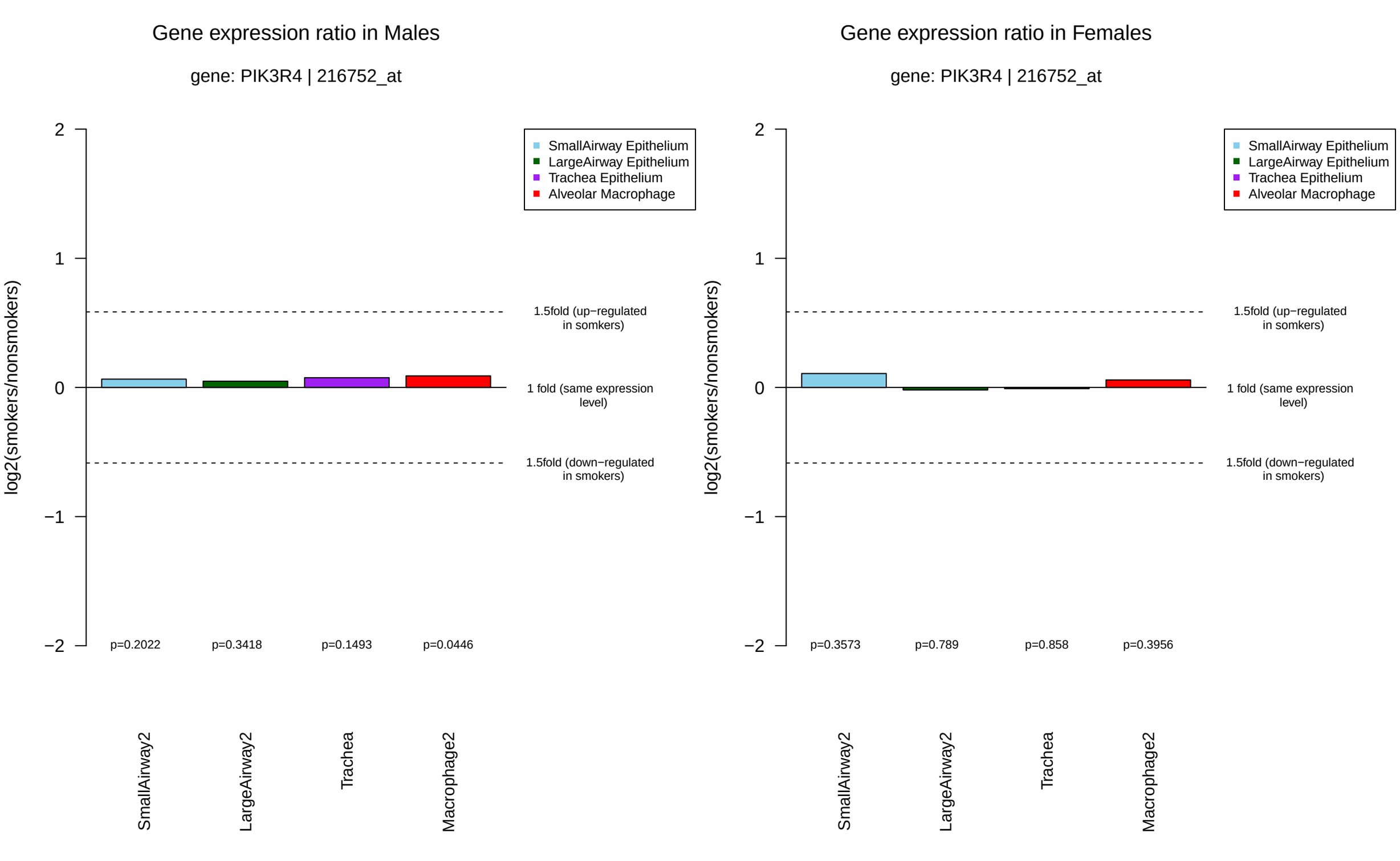
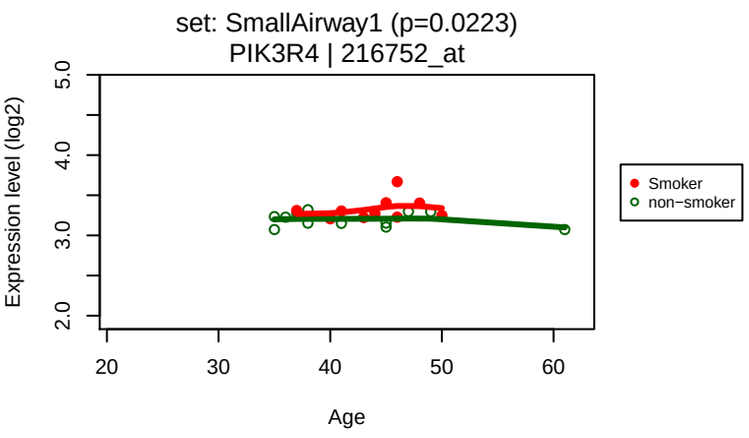
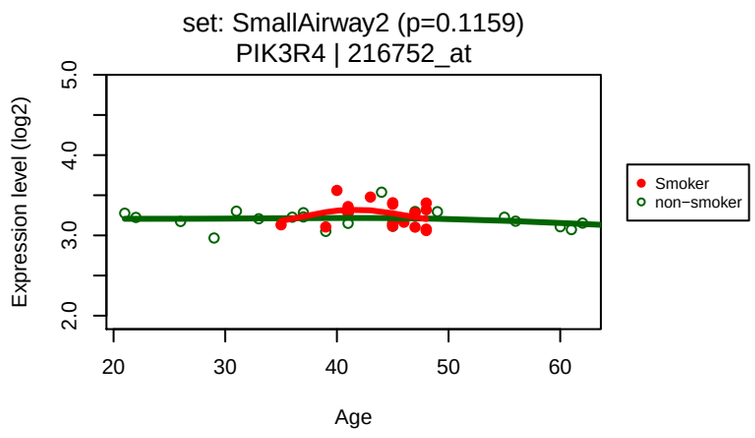
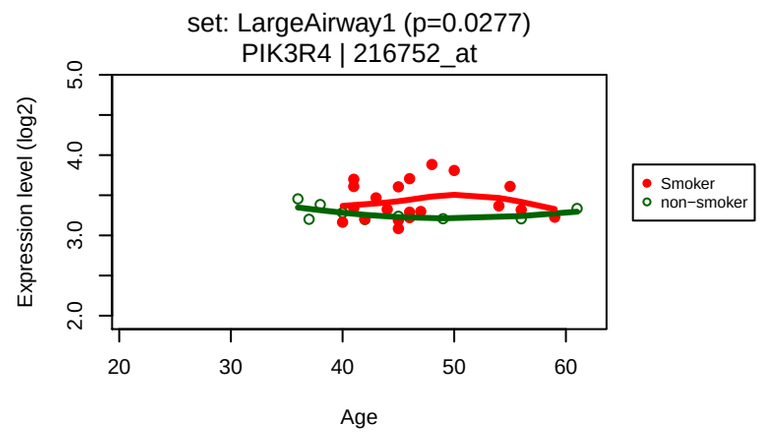

Gene expression of smokers and nonsmokers
gene: PIK3R4 | 216752_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0223
p=0.1159
p=0.0024
p=0.0277
p=0.6644
p=0.1264
p=0.6257
p=0.0248
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

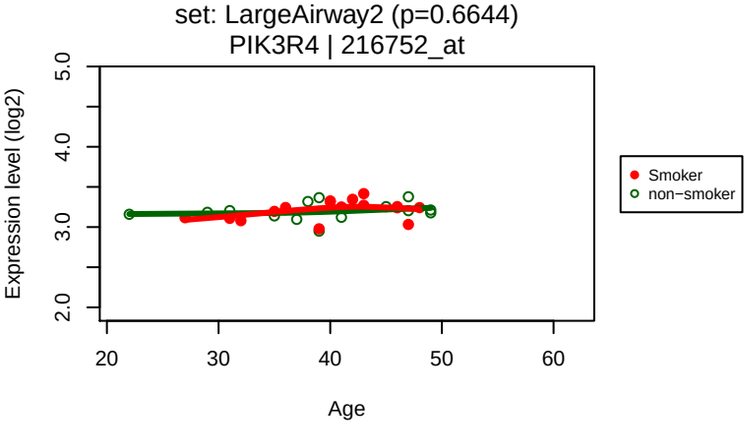

Gene expression ratio (smokers/nonsmokers)
gene: PIK3R4 | 216752_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0223
p=0.1159
p=0.0024
p=0.0277
p=0.6644
p=0.1264
p=0.6257
p=0.0248
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

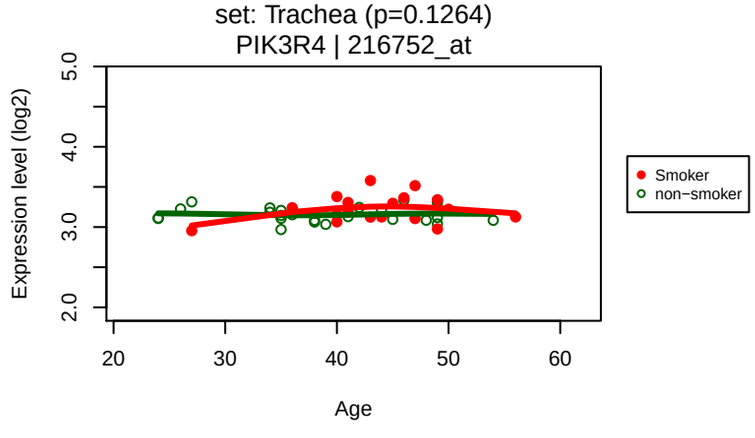

Gene expression ratio in Males
gene: PIK3R4 | 216752_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2022
p=0.3418
p=0.1493
p=0.0446
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: PIK3R4 | 216752_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3573
p=0.789
p=0.858
p=0.3956
SmallAirway2
LargeAirway2
Trachea
Macrophage2

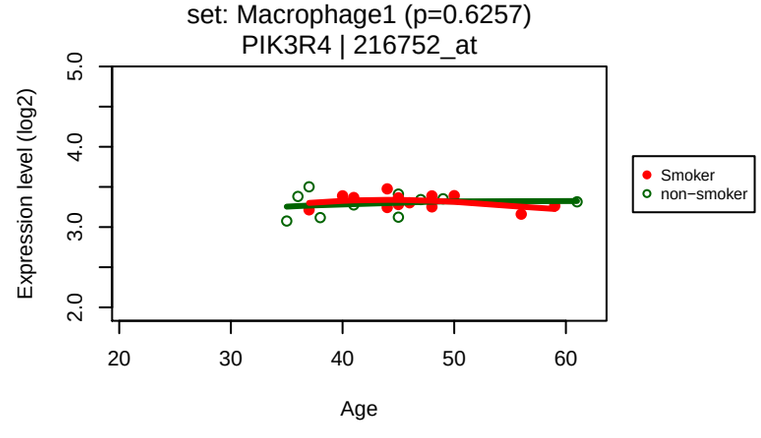

set: SmallAirway1 (p=0.0223)
PIK3R4 | 216752_at
Expression level (log2)
Age
Smoker
non−smoker

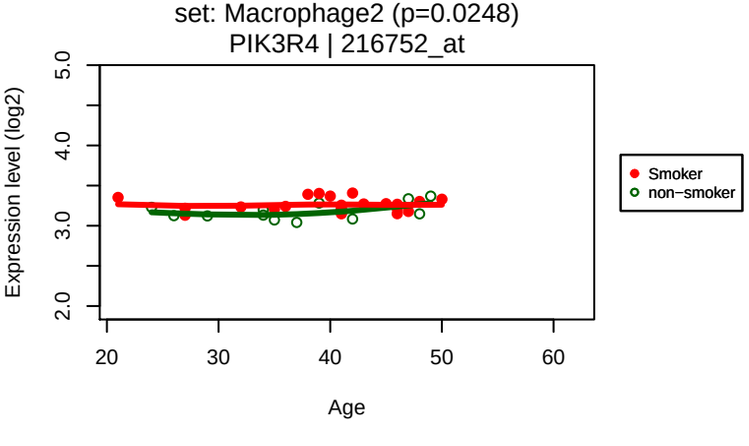

set: SmallAirway2 (p=0.1159)
PIK3R4 | 216752_at
Expression level (log2)
Age
Smoker
non−smoker

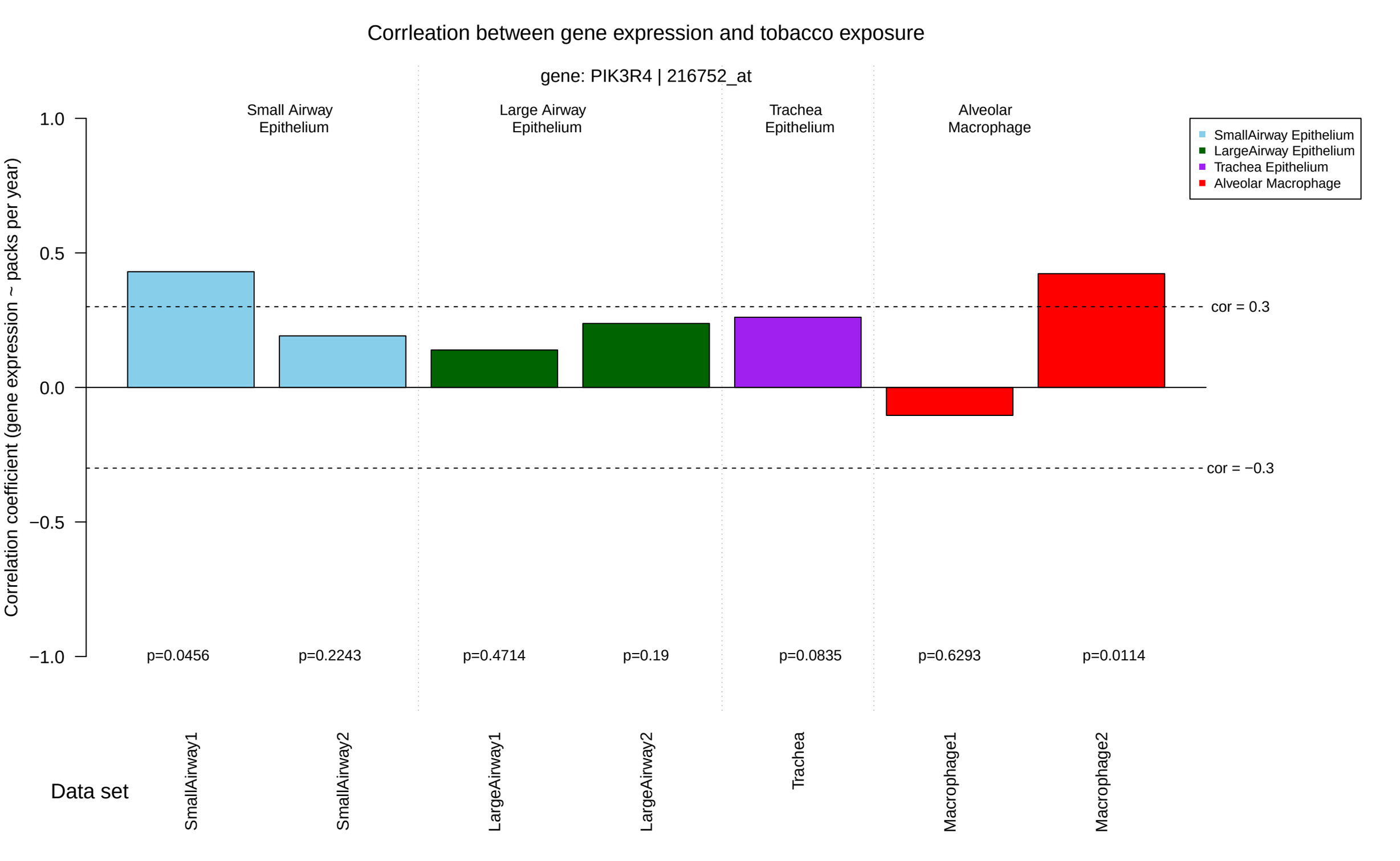

set: LargeAirway1 (p=0.0277)
PIK3R4 | 216752_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.6644)
PIK3R4 | 216752_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.1264)
PIK3R4 | 216752_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.6257)
PIK3R4 | 216752_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.0248)
PIK3R4 | 216752_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: PIK3R4 | 216752_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.0456
p=0.2243
p=0.4714
p=0.19
p=0.0835
p=0.6293
p=0.0114
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2